



# UNIVERSITY OF ASIA PACIFIC

## DEPARTMENT OF CSE

**Course Code: CSE 402**

**Course Title: Operating System Lab**

**Experiment No: 1**

**Experiment Title: Implementation of FCFS CPU Scheduling Algorithm**

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**Section : C-1**

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## Problem Statement:

Given the following processes with their Arrival Time (AT) and Burst Time (BT):

| Process | AT | BT |
|---------|----|----|
| P1      | 3  | 5  |
| P2      | 2  | 4  |
| P3      | 4  | 3  |
| P4      | 1  | 2  |
| P5      | 5  | 3  |

Implement **FCFS(Non-Preemptive)** scheduling algorithms. Calculate **CT, TAT, WT, Average TAT, and Average WT**, represent the execution sequence, and compare both algorithms to determine which performs better.

## Source Code:

```
processes =  
[ ["P1", 3, 5],  
  ["P2", 2, 4],  
  ["P3", 4, 3],  
  ["P4", 1, 2],  
  ["P5", 5, 3]  
]  
  
processes.sort(key=lambda x: x[1])  
  
time = 0  
total_tat = 0  
total_wt = 0  
print("PID\tAT\tBT\tCT\tAT\tWT")  
  
for pid, at, bt in processes:  
    if time < at:  
        time = at  
  
    time = time + bt  
    ct = time  
  
    tat = ct - at
```

```
wt = tat - bt
```

```
total_tat += tat
```

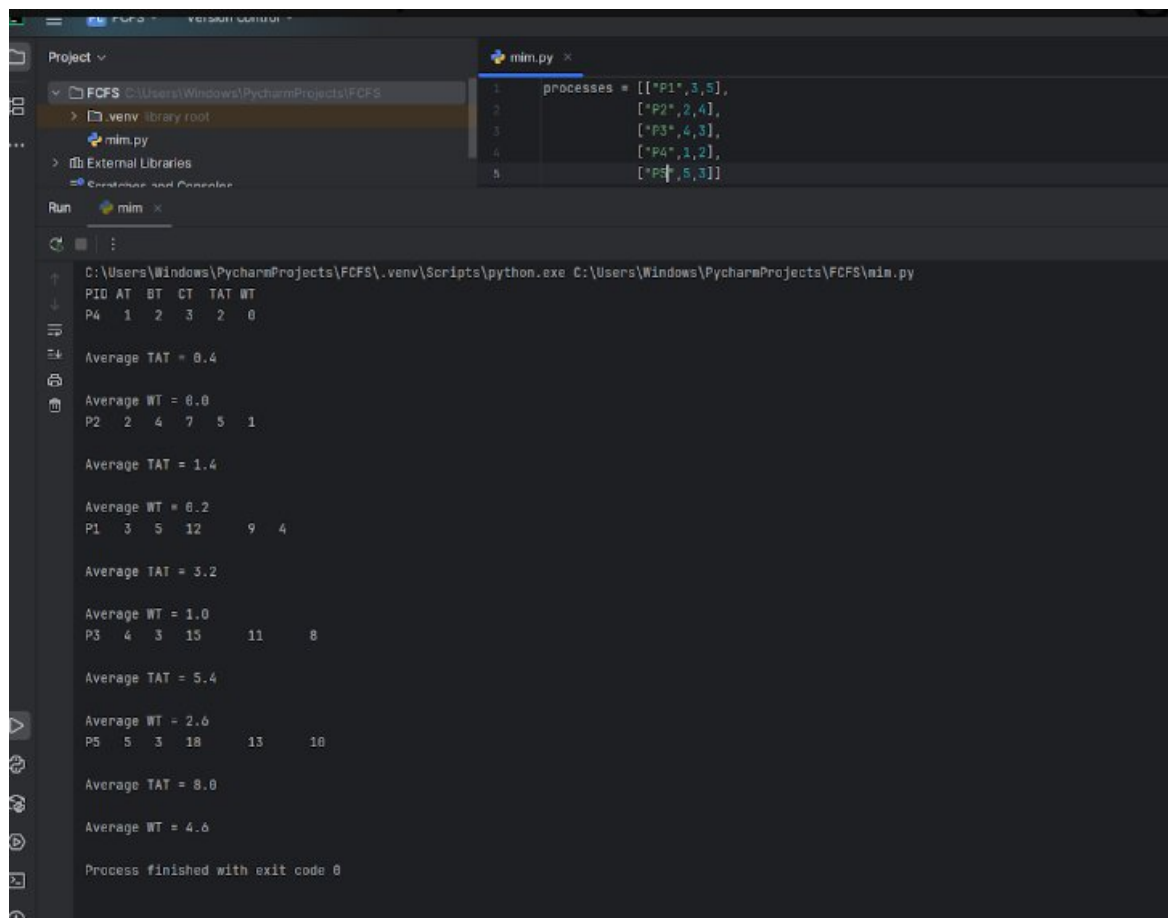
```
total_wt += wt
```

```
print(pid, "\t", at, "\t", bt, "\t", ct, "\t", tat, "\t", wt)
```

```
ava_tat = total_tat / len(processes)  
ava_wt = total_wt / len(processes)
```

```
print("\nAverage TAT =", ava_tat)  
print("Average WT =", ava_wt)
```

## Output:



```
min.py x
1 processes = [{"P1", 3, 5},
2               ["P2", 2, 4],
3               ["P3", 4, 3],
4               ["P4", 1, 2],
5               ["P5", 5, 3]]

C:\Users\Windows\PycharmProjects\FCFS\.venv\Scripts\python.exe C:\Users\Windows\PycharmProjects\FCFS\min.py
PID AT BT CT TAT WT
P4 1 2 3 2 0

Average TAT = 0.4

Average WT = 0.0
P2 2 4 7 5 1

Average TAT = 1.4

Average WT = 0.2
P1 3 5 12 9 4

Average TAT = 3.2

Average WT = 1.0
P3 4 3 15 11 8

Average TAT = 5.4

Average WT = 2.6
P5 5 3 18 13 10

Average TAT = 8.0

Average WT = 4.6

Process finished with exit code 0
```

## **Discussion:**

This experiment implements the FCFS CPU scheduling algorithm, where processes are executed strictly in their order of arrival. Using the given Arrival Time and Burst Time for each process, the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) were derived. The Average TAT and Average WT were then computed to assess overall performance. The results confirm that scheduling order has a direct impact on these metrics, since different scheduling algorithms select the next process to run using different criteria.

## **Conclusion:**

This experiment successfully demonstrates the working of the FCFS scheduling algorithm. Being non-preemptive and based purely on arrival order, FCFS is simple to implement and analyze. The computed Average TAT and Average WT values allow the efficiency of this execution order to be evaluated. Overall, the experiment illustrates how the choice of CPU scheduling strategy affects system performance.

## **GitHub:**

[https://github.com/23101134Mim/CSE\\_402\\_Operating\\_System\\_Lab](https://github.com/23101134Mim/CSE_402_Operating_System_Lab)